📦 Unit 9: Volume of 3D Solids

ANSWER KEY

Cylinders · Cones · Spheres · Composite Solids · NC.8.G.9

🎯 By the end of this unit, I can…

- Find the volume of cylinders, cones, and spheres
- Explain why a cone is one-third of its matching cylinder
- Solve real-world and composite-solid volume problems

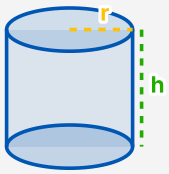
Words to know

- **Volume** — how much space a 3D shape holds. Units are cubed (cm^3).
- **Radius (r)** — center to edge of a circle.
- **Diameter (d)** — all the way across a circle. d = 2r.
- **Cylinder, cone, sphere** — a can, an ice-cream cone, a ball.
- **Hemisphere** — half of a sphere.
- **Composite solid** — a shape made of two or more simple shapes.

Key Formulas

🔑 **All volume formulas use π (pi ≈ 3.14).** On the EOG you can leave answers in terms of π (like 250π) **or** use 3.14. Using π or 3.14 gives answers that are a tiny bit different. Both are OK. The decimals below use 3.14. Always include **cubic units** (cm^3, in^3, ft^3, etc.).

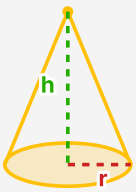
🥫 Cylinder



$$V = \pi r^2 h$$

Base area × height

🍦 Cone



$$V = \tfrac{1}{3}\pi r^2 h$$

One-third of a cylinder!

🏀 Sphere



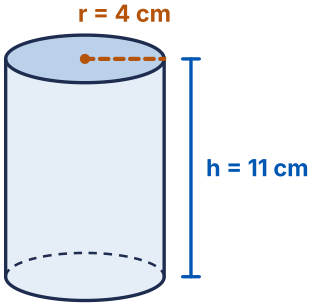
$$V = \tfrac{4}{3}\pi r^3$$

Only needs the radius!

✏ Practice: Pick the Formula

1. ★ Try it This soup can is a cylinder. Which formula finds its volume?

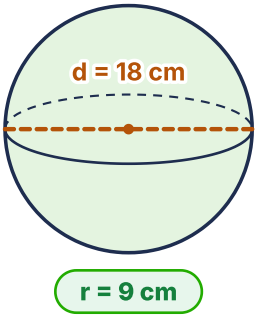
- ● **$V = \pi r^2 h$**
- ○ $V = \tfrac{1}{3}\pi r^2 h$
- ○ $V = \tfrac{4}{3}\pi r^3$



How to do it: A can has two circle bases and straight sides, so it is a cylinder. Use **$V = \pi r^2 h$** (base area × height).

2. ★★ On level This ball has a diameter of 18 cm. What radius goes into $V = \tfrac{4}{3}\pi r^3$? Write just the number.

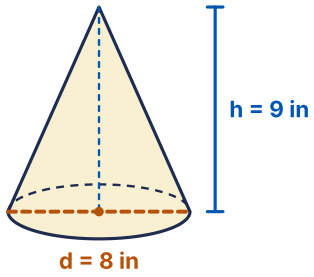
Answer: **r = 9 cm**



How to do it: r = d ÷ 2 = 18 ÷ 2 = **9 cm**. So r = 9 goes into the formula.

3. ★★★ Challenge This cone has a diameter of 8 in and a height of 9 in. Which setup is correct?

- ● **$V = \frac{1}{3}\pi(4)^2(9)$**
- ○ $V = \pi(4)^2(9)$
- ○ $V = \frac{1}{3}\pi(8)^2(9)$
- ○ $V = \frac{1}{3}\pi(9)^2(4)$

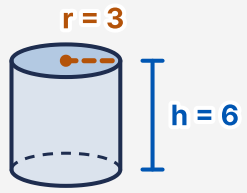


How to do it: Cone → $V = \frac{1}{3}\pi r^2 h$. Diameter 8 → r = 4. Height = 9. So **$V = \frac{1}{3}\pi(4)^2(9)$**. (It works out to 48π in³.)

Try It: Change r and h

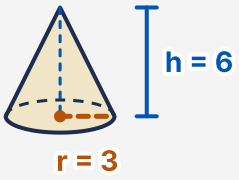
Here r = 3 and h = 6 for all three solids. Compare their volumes. Notice: the cone is always ⅓ of the cylinder. The sphere only cares about **r**.

radius r = **3** height h = **6**



Cylinder

$\pi(3)^2(6) = 54\pi \approx 169.6$



Cone

$\frac{1}{3}\pi(3)^2(6) = 18\pi \approx 56.5$



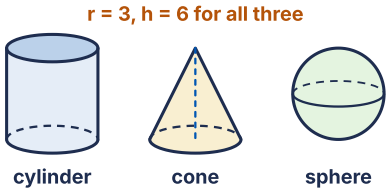
Sphere (uses r only)

$\frac{4}{3}\pi(3)^3 = 36\pi \approx 113.0$

✏ Practice: What Changes the Volume?

1. ★ Try it Which solid's volume formula does NOT use the height h?

- ○ Cylinder
- ○ Cone
- ● **Sphere**

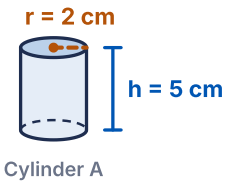


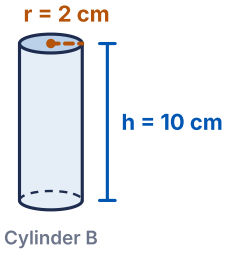
How to do it: Cylinder ($\pi r^2 h$) and cone ($\frac{1}{3}\pi r^2 h$) both use h. The **sphere** ($\frac{4}{3}\pi r^3$) uses only r.

2. ★★ On level Cylinder B is twice as tall as cylinder A. Same radius. How many times bigger is B's volume? Write just the number.

Answer: 2 times

How to do it: A: $\pi(2)^2(5) = 20\pi$. B: $\pi(2)^2(10) = 40\pi$. $40\pi \div 20\pi = \mathbf{2}$. Double the height → double the volume.

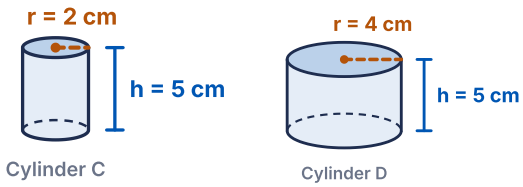




3. ★★★ Challenge Cylinder D has double the radius of cylinder C. Same height. How many times bigger is D's volume? Write just the number.

Answer: 4 times

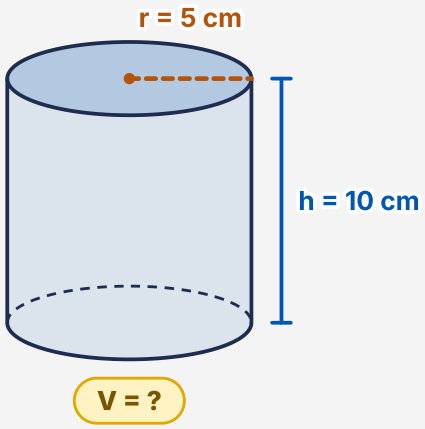
How to do it: C: $\pi(2)^2(5) = 20\pi$. D: $\pi(4)^2(5) = 80\pi$. $80\pi \div 20\pi = \mathbf{4}$. Double the radius → 4 times the volume, because r is squared ($2 \times 2 = 4$).



Worked Examples: 🥫 Cylinders

$V = \pi r^2 h$. Square the radius, multiply by π, then multiply by the height.

🥫 Cylinder: r = 5 cm, h = 10 cm



$V = \pi r^2 h$

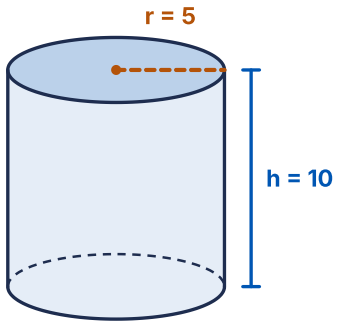
$V = \pi(5)^2(10)$

$V = \pi(25)(10)$

$V = 250\pi$

$V \approx 785.0$ cm³ (using 3.14; a calculator gives ≈ 785.4)

🔍 Spot the Error: Cylinder: r = 5, h = 10



× Student: V = π(5)(10) = 50π ≈ 157.0

✓ Fix: Forgot to SQUARE the radius! V = π(5²)(10) = π(25)(10) = 250π ≈ 785.0

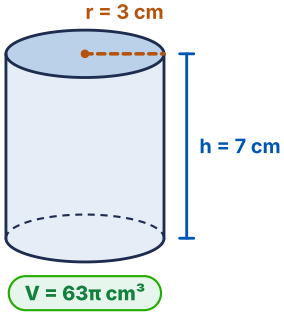
💡 The formula is πr²h — that ² on the radius is critical!

✏ Practice: Cylinders

1. ★ Try it Find the volume. Leave your answer in terms of π.

Answer: 63π cm³

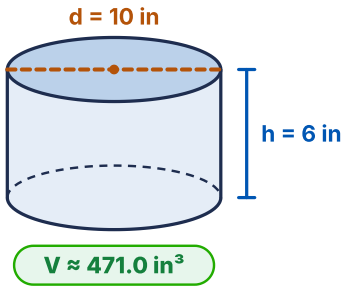
How to do it: V = πr²h = π(3)²(7) = π(9)(7) = **63π cm³**.



2. ★★ On level Find the volume. Use 3.14 and round to the nearest tenth.

Answer: ≈ 471.0 in³ (150π)

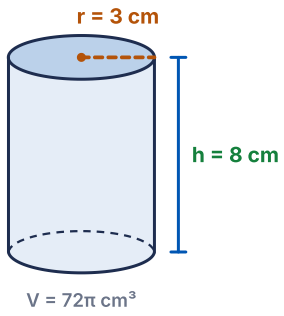
How to do it: r = 10 ÷ 2 = 5. V = π(5)²(6) = π(25)(6) = 150π ≈ 3.14 × 150 = **471.0 in³**.



3. ★★★ Challenge This can holds 72π cm³. Its radius is 3 cm. How tall is it? Write just the number.

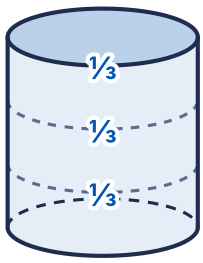
Answer: h = 8 cm

How to do it: 72π = π(3)²h → 72π = 9πh → h = 72 ÷ 9 = **8 cm**.

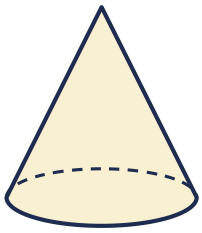


Relationship: Cylinder, Cone & Sphere

🔑 **THE BIG RELATIONSHIP:** A cone is exactly ⅓ of a cylinder with the same radius and height. So a cone's volume is always ⅓ of the matching cylinder.

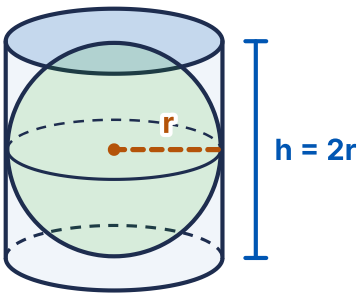


cylinder



cone (same r and h)

Fill the cone with water and pour it into the cylinder. It takes **3 cones** to fill it. So one cone = ⅓ of the cylinder.



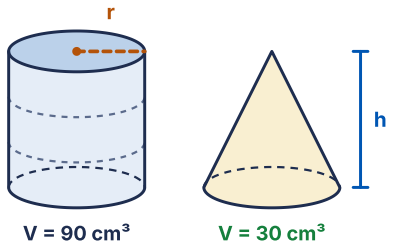
A sphere that just fits inside a cylinder (the height is 2r) fills ⅔ of it.

⭐ **Fun fact:** A ball that just fits in a can fills ⅔ of the can.

✏ Practice: Cone, Cylinder & Sphere

1. ★ Try it — The cylinder holds 90 cm³. The cone has the SAME radius and height. What is the cone's volume?

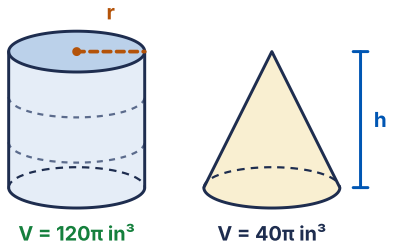
- ● **30 cm³**
- ○ 45 cm³
- ○ 90 cm³
- ○ 270 cm³



How to do it: A cone is always ⅓ of the matching cylinder: 90 ÷ 3 = **30 cm³**.

2. ★★ On level — The cone holds 40π in³. The cylinder has the same radius and height. Find the cylinder's volume. Leave it in terms of π.

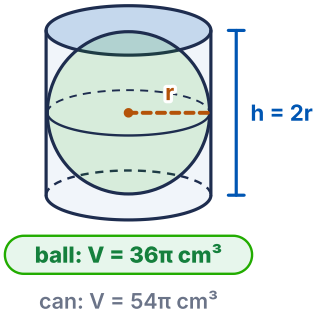
Answer: 120π in³



How to do it: Cylinder = 3 × cone = 3 × 40π = **120π in³**.

3. ★★★ Challenge — A ball fits exactly inside a can (h = 2r). The can holds 54π cm³. Find the ball's volume. Leave it in terms of π.

Answer: 36π cm³

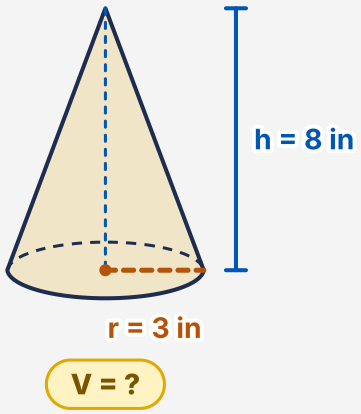


How to do it: Ball = ⅔ × can = ⅔ × 54π. 54π ÷ 3 = 18π, and 18π × 2 = **36π cm³**. (Check: this can has r = 3, h = 6, and ⁴⁄₃π(3)³ = 36π ✓)

Worked Examples: 🍦 Cones

$V = \frac{1}{3}\pi r^2 h$. Work it like a cylinder, then take ⅓ (divide by 3).

🍦 Cone: r = 3 in, h = 8 in



$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3} \cdot \pi(3)^2(8)$

$V = \frac{1}{3} \cdot \pi(9)(8)$

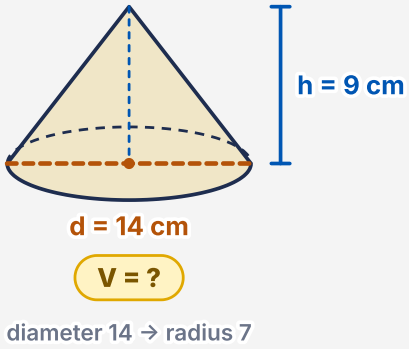
$V = \frac{1}{3} \cdot 72\pi$

$V = 24\pi$

$V \approx 75.4\ in^3$

📝 Worked Example

Volume of a cone: diameter = 14 cm, height = 9 cm



diameter 14 → radius 7

Step 1: Diameter = 14, so r = 7. h = 9

Step 2: Formula: $V = \frac{1}{3}\pi r^2 h$

Step 3: Substitute: $V = \frac{1}{3}\pi(7^2)(9) = \frac{1}{3}\pi(49)(9)$

Step 4: Multiply: $V = \frac{1}{3}(441\pi)$

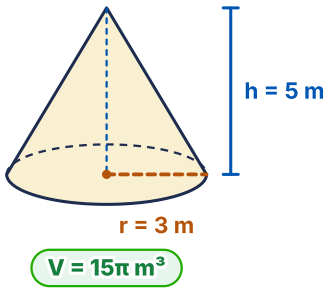
Step 5: $V = 147\pi \approx 461.6\ cm^3$ ✓

✏ Practice: Cones

1. ★ Try it Find the volume. Leave your answer in terms of π.

Answer: 15π m³

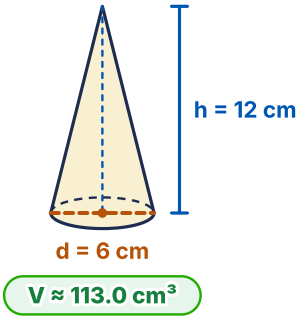
How to do it: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(3)^2(5) = \frac{1}{3}\pi(9)(5) = \frac{1}{3}(45\pi) =$ **15π m³**.



2. ★★ On level This cone has a diameter of 6 cm. Find its volume. Use 3.14 and round to the nearest tenth.

Answer: ≈ 113.0 cm³ (36π)

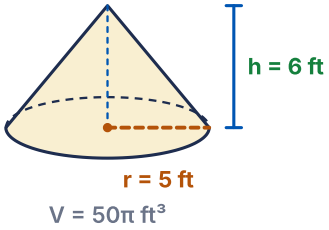
How to do it: r = 6 ÷ 2 = 3. $V = \frac{1}{3}\pi(3)^2(12) = \frac{1}{3}\pi(9)(12) = \frac{1}{3}(108\pi) = 36\pi \approx 3.14 \times 36 =$ **113.0 cm³**.



3. ★★★ Challenge This cone holds 50π ft³. Its radius is 5 ft. Find its height. Write just the number.

Answer: h = 6 ft

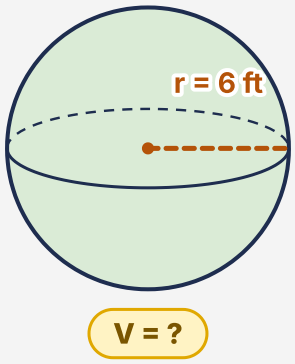
How to do it: $50\pi = \frac{1}{3}\pi(5)^2 h$. Drop the π on both sides: $50 = \frac{1}{3}(25)h$. Multiply by 3: 50 × 3 = 150, so 150 = 25h. Divide: 150 ÷ 25 = **6 ft**.



Worked Examples: 🏀 Spheres

$V = \frac{4}{3}\pi r^3$. CUBE the radius (r × r × r), then multiply by $\frac{4}{3}$ and π.

🏀 Sphere: r = 6 ft



$V = \frac{4}{3}\pi r^3$

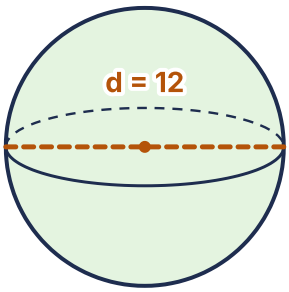
$V = \frac{4}{3} \cdot \pi(6)^3$

$V = \frac{4}{3} \cdot \pi(216)$

$V = \frac{4}{3} \cdot 216\pi = 288\pi$

$V \approx 904.3$ ft³ (using 3.14; a calculator gives ≈ 904.8)

🔍 Spot the Error: Sphere: diameter = 12



the 12 goes all the way across

× Student: r = 12, $V = \frac{4}{3}\pi(12^3) = \frac{4}{3}\pi(1728) \approx 7238.2$

✓ Fix: Used DIAMETER instead of radius! r = 6, not 12. $V = \frac{4}{3}\pi(6^3) = \frac{4}{3}\pi(216) = 288\pi \approx 904.8$

💡 Always check: did the problem say diameter or radius? Diameter ÷ 2 = radius.

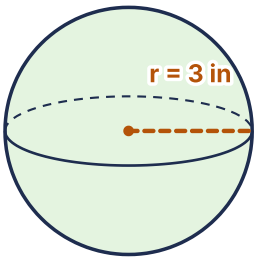
🔑 **HEMISPHERE = half a sphere** (like a bowl or a dome). Find the whole sphere, then divide by 2: $V = \frac{1}{2} \cdot \frac{4}{3}\pi r^3$.

✏ Practice: Spheres

1. ★ Try it Find the volume. Leave your answer in terms of π.

Answer: 36π in³

How to do it: V = ⁴⁄₃π(3)³ = ⁴⁄₃π(27) = **36π in³**. (27 ÷ 3 = 9, and 9 × 4 = 36.)

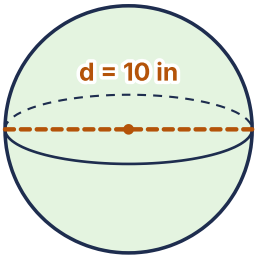


V = 36π in³

2. ★★ On level A ball has a diameter of 10 in. Find its volume. Use 3.14 and round to the nearest tenth.

Answer: ≈ 523.3 in³

How to do it: r = 10 ÷ 2 = 5. V = ⁴⁄₃π(5)³ = ⁴⁄₃π(125) = $\frac{500\pi}{3}$ ≈ 4 × 125 × 3.14 ÷ 3 = **523.3 in³**.

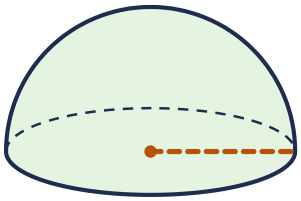


V ≈ 523.3 in³

3. ★★★ Challenge This bowl is a hemisphere with radius 6 cm. Find its volume. Leave your answer in terms of π.

Answer: 144π cm³

How to do it: Whole sphere: ⁴⁄₃π(6)³ = ⁴⁄₃π(216) = 288π. Half: 288π ÷ 2 = **144π cm³**.



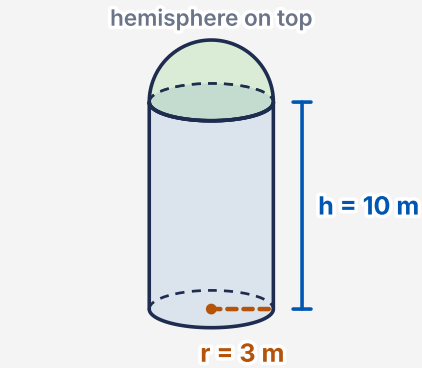
r = 6 cm

V = 144π cm³

Worked Examples: 📦 Composite & Real-World

A **composite solid** is made of simple solids. Find the volume of each part. Then add them.

📦 Composite: Cylinder + Hemisphere (r = 3 m, h = 10 m)



V = cylinder + half sphere

Cylinder: $V = \pi(3)^2(10) = \pi(9)(10) = 90\pi$

Full sphere: $V = \frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi(27) = 36\pi$

Hemisphere = half: $36\pi \div 2 = 18\pi$

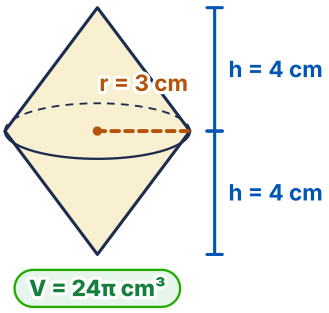
Total = $90\pi + 18\pi = 108\pi \approx 339.1$ m³ (using 3.14; a calculator gives ≈ 339.3)

✏ Practice: Composite & Real-World

1. ★ Try it — Two equal cones are glued base to base. Find the total volume. Leave your answer in terms of π.

Answer: 24π cm³

How to do it: One cone: $\frac{1}{3}\pi(3)^2(4) = \frac{1}{3}\pi(9)(4) = 12\pi$. Two cones: $2 \times 12\pi =$ **24π cm³**.

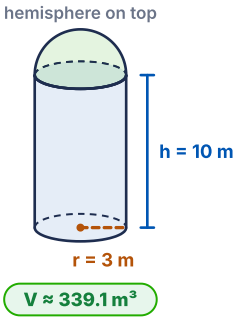


V = 24π cm³

2. ★★ On level — A grain silo is a cylinder with a hemisphere on top. Find its volume. Use 3.14 and round to the nearest tenth.

Answer: ≈ 339.1 m³ (108π)

How to do it: Cylinder: $\pi(3)^2(10) = 90\pi$. Hemisphere: $\frac{1}{2} \cdot \frac{4}{3}\pi(3)^3 = \frac{1}{2} \cdot 36\pi = 18\pi$. Total: $90\pi + 18\pi = 108\pi \approx 3.14 \times 108 =$ **339.1 m³**.

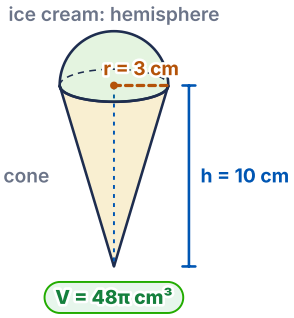


V ≈ 339.1 m³

3. ★★★ Challenge An ice cream cone is filled, with a hemisphere of ice cream on top. Find the total volume. Leave your answer in terms of π.

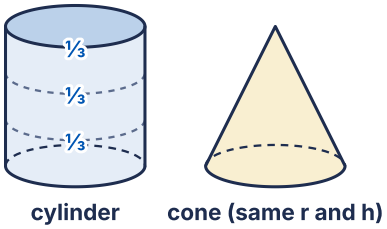
Answer: $48\pi\ \text{cm}^3$

How to do it: Cone: $\frac{1}{3}\pi(3)^2(10) = \frac{1}{3}\pi(90) = 30\pi$. Hemisphere: $\frac{1}{2} \cdot \frac{4}{3}\pi(3)^3 = \frac{1}{2} \cdot 36\pi = 18\pi$. Total: $30\pi + 18\pi = \mathbf{48\pi\ cm^3}$.

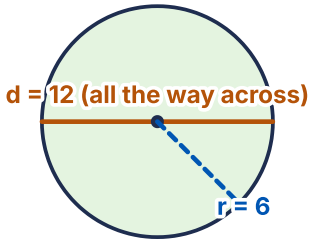


Common Mistakes

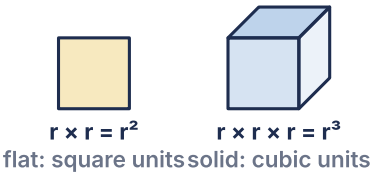
❌ **Forgetting ⅓ for cones** — The cone formula has ⅓. Without it, you get the cylinder's volume!



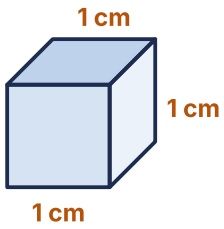
❌ **Using diameter instead of radius** — If the problem gives diameter, divide by 2 first!



❌ **Forgetting to cube for spheres** — The sphere uses r^3, not r^2. Square the radius for cylinders/cones, CUBE it for spheres.



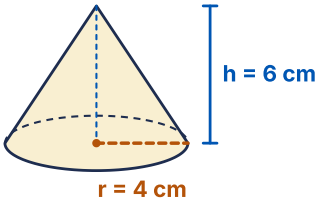
❌ **Missing units** — Volume is always in CUBIC units (cm^3, in^3, ft^3).



1 cm^3 = one cubic unit

✏ Practice: Spot the Mistake

1. ★ Try it A student found this cone's volume: $V = \pi(4)^2(6) = 96\pi$. What did they forget?

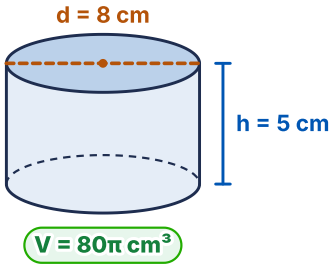


- ● **the ⅓**
- ○ to cube the radius
- ○ to cut the diameter in half

How to do it: Cones need **the ⅓**. Correct: $\frac{1}{3}\pi(4)^2(6) = \frac{1}{3}(96\pi) = 32\pi\ cm^3$. The 96π is the matching cylinder's volume.

2. ★★ On level Mia used r = 8 for this cylinder. Find the correct volume. Leave your answer in terms of π.

Answer: **$80\pi\ cm^3$**

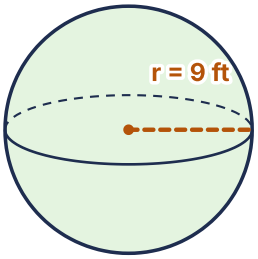


How to do it: r = 8 ÷ 2 = 4. $V = \pi(4)^2(5) = \pi(16)(5) =$ **$80\pi\ cm^3$**. (Mia's answer, $\pi(8)^2(5) = 320\pi$, is 4 times too big.)

3. ★★★ Challenge Sam wrote $V = \frac{4}{3}\pi(9)^2 = 108\pi$ for this sphere. Find the correct volume. Leave your answer in terms of π.

Answer: **972π ft³**

How to do it: Sam squared the radius. Spheres CUBE it: $V = \frac{4}{3}\pi(9)^3 = \frac{4}{3}\pi(729) =$ **972π ft³**. (729 ÷ 3 = 243, and 243 × 4 = 972.)



V = 972π ft³

Putting It All Together

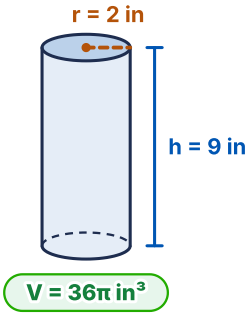
Mixed practice from every section. Read each one: which solid is it? Radius or diameter? 3.14 or π?

✏ Comprehensive Practice All sections

1. ★ Try it Find the volume of the cylinder. Leave your answer in terms of π.

Answer: 36π in³

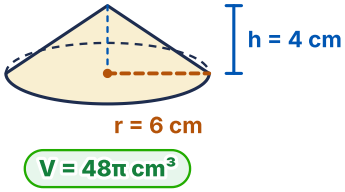
How to do it: V = π(2)²(9) = π(4)(9) = **36π in³**.



2. ★ Try it Find the volume of the cone. Leave your answer in terms of π.

Answer: 48π cm³

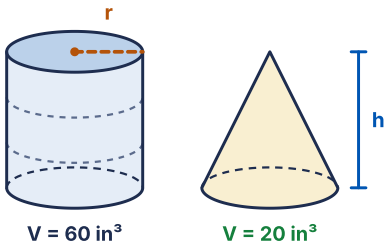
How to do it: V = ⅓π(6)²(4) = ⅓π(36)(4) = ⅓(144π) = **48π cm³**.



3. ★ Try it The cylinder holds 60 in³. The cone has the same radius and height. Find the cone's volume. Write just the number.

Answer: 20 in³

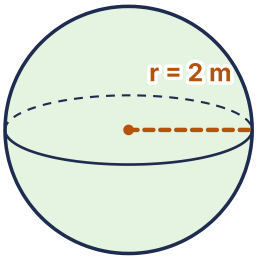
How to do it: Cone = ⅓ × cylinder = 60 ÷ 3 = **20 in³**.



4. ★★ On level Find the volume of the sphere. Use 3.14 and round to the nearest tenth.

Answer: ≈ 33.5 m³

How to do it: $V = \frac{4}{3}\pi(2)^3 = \frac{4}{3}\pi(8) = \frac{32\pi}{3} \approx 4 \times 8 \times 3.14 \div 3 \approx$ **33.5 m³**.

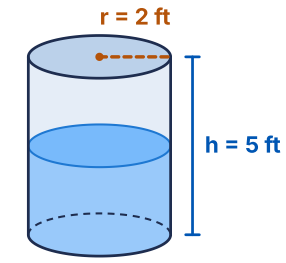


V ≈ 33.5 m³

5. ★★ On level This water tank is a cylinder. It is half full. How much water is in it? Leave your answer in terms of π.

Answer: 10π ft³

How to do it: Whole tank: $\pi(2)^2(5) = 20\pi$. Half full: $20\pi \div 2 =$ **10π ft³**.

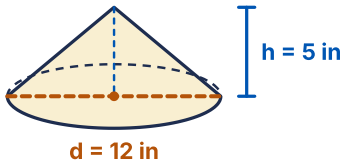


water = 10π ft³

6. ★★ On level Find the volume of the cone. Use 3.14 and round to the nearest tenth.

Answer: ≈ 188.4 in³ (60π)

How to do it: r = 12 ÷ 2 = 6. $V = \frac{1}{3}\pi(6)^2(5) = \frac{1}{3}\pi(180) = 60\pi \approx 3.14 \times 60 =$ **188.4 in³**.

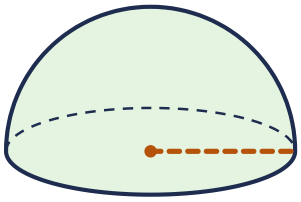


V ≈ 188.4 in³

7. ★★ On level Find the volume of the hemisphere. Leave your answer in terms of π.

Answer: 18π cm³

How to do it: Whole sphere: $\frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi(27) = 36\pi$. Half: $36\pi \div 2 =$ **18π cm³**.

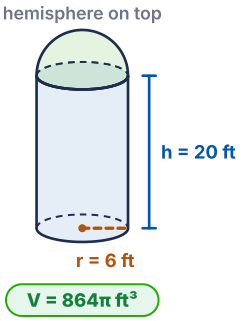


V = 18π cm³

8. ★★★ Challenge A silo is a cylinder with a hemisphere on top. Find its total volume. Leave your answer in terms of π.

Answer: **864π ft³**

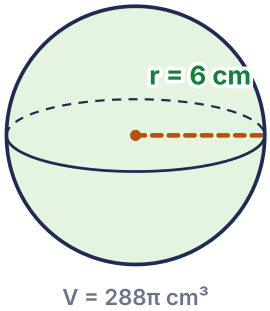
How to do it: Cylinder: $\pi(6)^2(20) = \pi(36)(20) = 720\pi$. Hemisphere: $\frac{1}{2} \cdot \frac{4}{3}\pi(216) = \frac{1}{2} \cdot 288\pi = 144\pi$. Total: $720\pi + 144\pi =$ **864π ft³**.



9. ★★★ Challenge A sphere holds 288π cm³. Find its radius. Write just the number.

Answer: **r = 6 cm**

How to do it: $288\pi = \frac{4}{3}\pi r^3 \rightarrow 288 \times 3 = 864 \rightarrow 864 \div 4 = 216 \rightarrow r^3 = 216 \rightarrow r =$ **6 cm** ($6 \times 6 \times 6 = 216$).



10. ★★★ Challenge Which cup holds more: the cylinder cup or the cone cup?

- ○ the cylinder cup
- ○ the cone cup
- ● **they hold the same**

How to do it: Cylinder: $\pi(3)^2(4) = 36\pi$. Cone: $\frac{1}{3}\pi(3)^2(12) = \frac{1}{3}(108\pi) = 36\pi$. **They hold the same!** The cone is 3 times as tall, which makes up for the ⅓.

